MIL-F-17619E(NU)
28 October 1981
SUPERSEDING
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7 January 1975

MILITARY SPECIFICATION

FRAME, SERVICE CAP (MAN'S)

This specification is approved for use by the Navy Clothing and Textile Research Facility, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope - This specification covers the requirements for cap frames, to be worn by male personnel of the Navy and Coast Guard.

*1.2 Classification - The cap frames shall be procured in the following types, classes, styles, and sizes as specified (see 6.2).

Type I - Frame, with cloth top visor

Class 1 - For Flag Rank Officers
Class 2 - For Captains and Commanders

Type II - Frame, with plastic top visor

Class 1 - With 2 silver buttons (NAVY)
Class 3 - With 2 gold buttons (CG)
Class 4 - With 2 gold buttons (NAVY)

Style A - Front stand with removable crown support wire
Style B - Front stand with non-removable crown support wire

Schedule of sizes

6 1/2	7 1/4
6 5/8	7 3/8
6 3/4	7 1/2
6 7/8	7 5/8
7	7 3/4
7 1/8	7 7/8
	8

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing & Textile Research Facility, 21 Strathmore Road, Natick, MA 01760 by using the self-addressed standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8405

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2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents - The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-390 - Plastic, Molding and Extrusion Material, Polyethylene and Copolymers (Low, Medium, and High Density)
V-T-276 - Thread, Cotton
V-T-295 - Thread, Nylon
DDD-L-20 - Label; For Clothing, Equipage, and Tentage (General Use)
PPP-B-636 - Boxes, Shipping, Fiberboard

MILITARY

MIL-B-3461 - Buttons, Insignia, Metal, Uniform and Cap
MIL-S-3577 - Sweatband, Headwear, Leather
MIL-C-16290 - Cloth, Melton, Wool (Moth-Proofed)
MIL-E-17568 - Embroidery Materials, Metallic and Synthetic Metallic
MIL-L-43541 - Leather, Cattlehide, for Visors and Chinstraps
MIL-T-43548 - Thread, Polyester, Cotton, Covered
MIL-S-43993 - Sweatband, Headwear, Artificial Leather
MIL-C-82252 - Cloth, Broadcloth, Wool, and Wool-Synthetic (Moth-Proofed)

STANDARDS

FEDERAL

FED-STD-151 - Metals, Test Methods
FED-STD-191 - Textile Test Methods
FED-STD-311 - Leather, Methods of Sampling and Testing
FED-STD-595 - Colors
FED-STD-751 - Stitches, Seams and Stitchings

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

LAWS AND REGULATIONS

U.S. POSTAL SERVICE MANUAL

(Copies of the manual may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C. 20402.)

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications - The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT
National Motor Freight Classification

(Application for copies should be addressed to American Trucking Association, Inc., ATTN: Traffic Department, 1616 P Street, N.W., Washington, D. C. 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT
Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illi[illegible].)

3. REQUIREMENTS

3.1 Guide sample - Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from this specification may appear in the sample, in which case this specification shall govern.

* 3.2 First article - When specified (See 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (See 4...

3.3 Materials -

* 3.3.1 Cloth, wool or wool/synthetic - The cloth used for the type I cap frame visor top and for covering the bottom edge of the inner band on type I and II cap frames shall conform to type I, class 2 of MIL-C-16290 or type I, class III or type II, class 3 of MIL-C-82252.

* 3.3.2 Front stand - The front stand shall be made of a high density (linear) polyethylene plastic material, white, natural or black in color, conforming to type I, II or III, class H, grades 1 or 2 of L-P-390, except that the low temperature brittleness, dielectric constant, dissipation factor, and thermal stress cracking requirements shall not be applicable. The front stand shall be molded into a shape similar to those shown in figure 3. The minimum thickness of the plastic material shall be 0.050 inch. Testing shall be as specified in 4.3.1.

3.3.2.1 Stiffness - The front stand material shall have a minimum stiffness scale rating of 55 when tested as specified in 4.3.1.

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3.3.2.2 Bending - The front stand material shall show no evidence of cracking when tested as specified in 4.3.1.

3.3.3 Cloth, cotton twill - The cloth, unbleached, shall be suitable for laminating to the wool broadcloth or melton as specified in 3.3.1, and shall be used on the type I cloth visor top. The cloth shall have a minimum of 72 ends and 60 picks per inch, weigh 6 to 8 ounces per square yard, and shall have a maximum of 15 percent starch and protein content including chloroform and water-soluble material when tested as specified in 4.3.1.

* 3.3.4 Plumper paper - A suitable commercial plumper paper shall be used for laminating to the plastic sheeting in the fabrication of the type II plastic visor top. The plumper paper shall be 0.020 ± 0.002 inch thick and properly treated so that it retains its original shape after being subjected to the immersion test specified in 4.3.1.

3.3.5 Plastic sheeting (non-rigid) - The plastic sheeting shall be used on the type II plastic visor top and for the visor binding on types I and II cap frames in accordance with the requirements specified in 3.3.5.1 and 3.3.5.2. The plastic sheeting shall be of a non-rigid porperly plasticized polyvinyl chloride or copolymer of vinyl-chloride-acetate, pigmented black approximating shade No. 17038 of FED-STD-595, and shall show no stiffening; tackiness; cracking; or flaking when tested as specified in 4.3.1.

3.3.5.1 Visor top - The plastic sheeting shall be press-polished on one side, and shall be 0.020 (± 0.002) inch in thickness when tested as specified in 4.3.1.

3.3.5.2 Visor binding - The binding shall be fabricated from plastic sheeting, [illegible] (± 0.003) inch in thickness and 3/4 to 7/8 inch in width when tested as specified in 4.3.1.

* 3.3.6 Leather - The leather used in fabricating the visor bottom of types I and II cap frame visors, shall conform to type II of MIL-L-43541; 6 to 7 ounces in thickness; vegetable and chrome combination tanned; except that the finish shall be of a boxed, diced, or pin-seal pattern. As an alternate, the visor bottom may be constructed of reconstituted leather (commercially referred to as PARLON). When reconstituted leather is used, the thickness shall be 6 to 7 ounces. The color of the visor bottom may be black or Hatter's green.

* 3.3.7 Adhesive - The adhesive used for bonding the component parts of the visor shall be water resistant and made from either natural or synthetic rubber. Testing shall be as specified in 4.3.1.

3.3.8 Embroidered visors (Type I frame) - When cloth top visors are to be embroidered, the embroidery materials shall conform to types I, II, or III of MIL-E-17568 (see figure 2).

* 3.3.9 Inner band - The inner band shall be made from a high density (linear) polyethylene in flat stock or molded form conforming to the requirements specified in 3.3.9.1 and 3.3.9.2.

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3.3.9.1 Standard inner band - The high density (linear) polyethylene plastic material shall be either opaque white, natural, or black, in color conforming to types I, II, or III, class H, grades 1 or 2, of L-P-390, except that the low temperature brittleness, dielectric constant, dissipation factor, and thermal stress cracking resistance requirements shall not be applicable. The standard inner band shall be made from flat stock that has been extruded, stripped, slit, or cut 2 (± 1/16) inches in width, 0.050 to 0.060 inch in thickness, and shall be perforated to match the Navy's reference sample (see 6.3). A margin of 3/8 inch at the top and bottom edges of the band material shall be unperforated. Testing shall be as specified in 4.3.1.

3.3.9.1.1 Stiffness - The standard inner band material shall have a stiffness (scale reading) between 11 and 27 inclusive when tested as specified in 4.3.1.

3.3.9.1.2 Bending - The standard inner band material shall show no evidence of cracking after the bending test specified in 4.3.1.

3.3.9.2 Alternate inner band - The alternate inner band shall be made from the material specified in 3.3.9.1, meet the requirements of 3.3.9.1.1 and 3.3.9.1.2, and shall be molded in a cane interlaced chair seat design on one or both sides. The top and bottom margins shall finish 3/16 to 1/4 inch in width and shall be 0.050 to 0.075 inch in thickness. The thickness of the material in the chair seat design area shall be approximately 0.010 inch less than that in the margins. The finished width of the inner band shall be 2 (± 1/8) inches. Testing shall be as specified in 4.3.1.

* 3.3.10 Sweatband - The sweatband shall be made from either sheepskin leather conforming to MIL-S-3577 or from artificial leather conforming to MIL-S-43993. The sweatband shall finish 1 1/2 to 1 3/4 inches in width and shall be black or brown in color. The finished sweatband shall contain perforations that are evenly spaced, except for a three inch long area at back end of sweatband which shall be left unperforated.

3.3.11 Sweatband welt - The material used for the sweatband welt shall be an unsupported flexible vinyl film, 0.010 inch minimum thickness and crack resistant at 0°F. The color of the film shall approximate the shade of the sweatband. Testing shall be as specified in 4.3.1.

3.3.12 Crown protector - The material used for the crown protector shall conform to one of the following:

(a) Vinyl coated cloth, white, smooth or embossed surface, 15 ounce minimum weight per square yard, knitted cotton or polyester base fabric, coating shall be crack resistant at minus 20°F. Testing shall be as specified in 4.3.1.

(b) Unsupported flexible vinyl sheeting, white, 0.010 inch minimum thickness, crack resistant at minus 20°F. Testing shall be as specified in 4.3.1.

3.3.13 Card holder - The material used for the card holder affixed to the face side of the crown protector shall be a semi-rigid, clear vinyl or cellulose acetate sheeting 0.004 to 0.008 inch in thickness. Testing shall be as specified in 4.3.1.

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3.3.14 Metal fittings -

3.3.14.1 Grommet wire - The grommet wire used to support the cap crown shall be made of first quality spring steel completely coated with white ethyl cellulose, cellulose propionate or cellulose acetate propionate coating. The steel wire shall be cut to the required length (see table II) and shall finish 5/16 (± 1/32) inch in width and 0.020 inch (± 0.005) inch in thickness. Testing shall be as specified in 4.3.1.

* 3.3.14.2 Clamp - The adjustment clamp clinched to one end of the steel grommet wire shall be made of the following materials:

Base and clamp	- Carbon steel AISI 1020, hardness Rockwell B scale 66, thickness 0.031 (± 0.002) inch.
Slug	- Carbon steel AISI 1015/1020, hardness Rockwell B scale 90, thickness 0.015 (± 0.002) inch
Assembled clamp	- Electroplated with zinc 0.00005 inch minimum thickness, chrome plated finish

The clamp assembly shall be approximately 1 inch long and its design shall conform to figure 4. The width of the base component shall be sufficient to accommodate the width of the grommet wire. Testing shall be performed as specified in 4.3.1.

* 3.3.14.3 Staples - The staples shall be stamped from a heavy duty corrosion resistant flat steel wire, 0.0250 (± 0.0005) inch in thickness, and shall be of sufficient length to thoroughly secure the applicable component parts and leave a smooth flat clinch without any distortion of the materials. The staples shall have a corrosion resistant finish, and shall show no evidence of corrosion when tested as specified in 4.3.1.

* 3.3.15 Buttons -

3.3.15.1 Navy - The buttons used for type II cap frames shall be 22 1/2 ligne conforming to MIL-B-3461, except that the length of the screw shall be 3/8 (± 1/64) inch. Silver buttons for the class 1 frame shall conform to type II, style 4, class C and gold buttons for the class 4 frame shall conform to type II, style 4, class A.

3.3.15.2 Coast Guard - The buttons used for the type II, class 3, cap frame shall be 22 1/2 ligne conforming to type II, style 5, class A, of MIL-B-3461, except that the length of the screw tube shall be 3/8 (± 1/64) inch.

3.3.16 Thread -

* 3.3.16.1 Thread, cotton - The cotton thread used for seaming and stitching the cap frame shall conform to type IA3 (soft); ticket no. 30/3, type IB3 (glazed) 4 ply, ticket no. 24; type IC1 (mercerized) 3 ply, ticket no. A and type IC2 (mercerized) 3 ply, ticket no. A and 00 of V-T-276. The color of the thread shall be bleached white or dyed black or brown to approximately match the material to which it is sewn. The dyed thread shall show good fastness to laundering, perspiration, wet-dry cleaning, and light.

Note: Cotton covered polyester threads conforming to MIL-T-43548, in equivalent sizes, may be substituted for the cotton threads.

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* 3.3.16.2 Thread, nylon - The nylon thread for seaming and stitching the sweatband assembly to the cap frame shall conform to class A, type I, II or III, size E or type IV, size C of V-T-295. The thread shall be dyed black or brown to approximate the shade of the sweatband and shall show good colorfastness to wet-dry cleaning and perspiration.

3.3.17 Labels - Each cap frame shall have a combination identification-size label, a separate size label, and an instruction label conforming to the requirements specified in 3.3.17.1, 3.3.17.2, and 3.3.17.3.

* 3.3.17.1 Identification-size label - The combination identification and size label shall conform to type III or IV, class 4 of DDD-L-20; except that the label shall bear the following inscription:

FRAME, SERVICE CAP (MAN'S)
CONTRACT NO.: DLA100-00-0-0000 (EXAMPLE)
STOCK NO.: 8405-00-000-0000 (EXAMPLE)
SIZE: 7 (EXAMPLE)
NAME OF CONTRACTOR:

The printing on the label shall show fastness to dry cleaning only.

3.3.17.2 Size label, stamped or paper - The stamped size label shall conform to type IV, class 2, of DDD-L-20; except that only the size shall be designated and shall show fastness to dry cleaning only. The paper label shall be of a pressure-sensitive adhesive type, with the size number printed in black with numerals not less than 5/16 inch in height.

3.3.17.3 Special instruction label - A special instruction label shall be packaged with the cap frame as specified in 5.1.1. The label shall measure not less than 2 inches by 3 inches. The text shall be printed on one side only, in letters not less than 3/32 inch in height. Inscription shall be as follows:

SPECIAL INSTRUCTIONS

IF THE CIRCUMFERENCE OF THE FRAME WIRE REQUIRES
ADJUSTMENT TO FIT THE CAP COVER, THE CLAMP SHALL BE
LOOSENED AND RE-SECURED TO THE PROPER SIZE.

* 3.4 Design - The cap frame shall have a circular inner band made of plastic, a molded plastic front stand and a crescent shaped visor made from combinations of wool, plastic and leather. A circular adjustable steel wire with attached vinyl crown protector shall be used to support the cap cover. Type II cap frames shall have screw type metal button fastened to each side of frame. Type I cap frames are supplied without buttons. The frames shall be so constructed that removable cap covers of corresponding sizes may fit snugly over them (see figure 1 thru 4).

3.5 Figures - Figures 1 thru 4 are furnished for information purposes only. When inconsistencies exist between the written specification and the figures, the written specification shall govern (see 6.4).

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3.6 Patterns - The patterns for cutting all parts of the cap frame shall be furnished by the contractor, and shall be of the proper proportions to provide good fitting cap frames conforming to the finished dimensions in Table II.

3.7 Construction -

3.7.1 Stitches, seams and stitching - Stitch, seam, and stitching types specified in Table I shall conform to FED-STD-751. Whenever two or more methods, seams, or stitches are given for the same operation, any one of them may be used. Where stitch type 101 or 401 is used, the chain or looper (underthread) shall be on the inside of the cap frame.

3.7.2 Stitches per inch - The minimum and maximum number of stitches per inch shall be as specified in Table I.

3.7.3 Thread breaks and ends of seams - Unless otherwise specified in Table I, ends of all seams and stitching, when not caught in other seams or stitching, shall be backtacked not less than 1/4 inch. The ends of a continuous line of stitching shall be overlapped not less than 1/2 inch. Thread breaks (all stitch types) shall be secured by starting the repair stitch not less than 1/2 inch in front of break. Stitchlines used on the circular sections of the cap frame must overlap 3/8 to 5/8 inch beyond the starting point or if necessary backstitched to prevent ravelling. When stitch type 101 or 401 is permitted as an option, it shall not be used unless the ends are caught in other seams or rows of stitching.

* 3.8 Manufacturing operations - The cap frames shall be manufactured in accordance with operation requirements specified in Table I. The contractor is not required to follow the exact sequence of operations, provided the finished cap frames are identical to those produced by following the exact sequence as shown.

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Table I - Construction of cap frames

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
1.	Cutting a. Cut the component parts of the cap frame in strict accordance with the contractor's patterns. b. Cut the wool broadcloth or wool melton for the welt strip around the bottom of the inner band 2 (± 1/8) inches in width (see 3.3.1). c. Cut the material for the crown protector piece approximately 4 1/2 inches in width at the center, then tapering the sides down to approximately 1 1/2 inches wide at each end. These crown protectors will vary in length, depending on cap size, so that they will not sag excessively in the finished cap (see 3.3.12 and figure 4). d. Cut the vinyl film for the sweatband welt 3/4 (± 1/16) inch in width (see 3.3.11). e. Cut the plastic sheeting (cellulose acetate or vinyl) for the card holder into strips 4 1/2 (± 1/8) inches in length by 3 1/4 (± 1/8) inches in width (see 3.3.13). f. Cut the grommet wire to the lengths specified in Table II (see 3.3.14.1). g. Cut a 2 inch wide by 6 1/2 inch long piece of white vinyl coated fabric for binding the top edge of style A front stand (see 3.3.12(a)).					
2.	Replacement of defective components During the cutting and manufacturing operations, components having material defects or damages that are classified as defects in Section 4, shall be removed from production and replaced with non-defective and properly sized components.					

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Table I Construction of cap frames (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
3.	Make visor The finished visor shall be as shown in Figure 2. a. The type I visor shall be constructed as follows: 1. Top piece - Laminate the wool broadcloth or melton (see 3.3.1) to the reinforcement piece of cloth, cotton twill (see 3.3.3) face side up with the adhesive specified in 3.3.7. To the wool broadcloth or melton, stitch a gold metallic applique (see 3.3.8 and Figure 2).	Embroidered				
	2. Assembly - Laminate the embroidered top piece to the leather bottom piece with the adhesive specified in 3.3.7. Skive the assembled components 1/4 (± 1/16) inch along the inside end from tip to tip. b. The type II visor shall be constructed as follows: 1. Top piece - Laminate the plastic sheeting (see 3.3.5) to the plumper paper (see 3.3.4) with the adhesive specified in 3.3.7. 2. Assembly - Laminate the top piece to the leather bottom piece with the adhesive specified in 3.3.7. Skive the assembled components 1/4 (± 1/16) inch along the inside end from tip to tip.					
	c. For types I and II visors, stitch the visor binding (see 3.3.5.2) around the outside edge 1/16 inch from the turned edge of binding on top of visor, and catching the other edge in the stitching on the bottom of the visor. The binding shall finish on the top of the visor 3/16 to 1/4 inch.	301	BSb-1	7-8	A/3	A/3

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Table I - Construction of cap frames (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
3.	Make visor (cont'd) d. The finished visors for types I and II cap frames shall be properly molded with heat into a uniform crescent shape, and shall be of one size only with dimensions as follows: 1. Overall width of visor before assembly, at center 2-3/4 (± 1/8) inches. 2. Width of visor after assembly, at center from front edge to start of skiving 2-3/8 (± 1/8) inches. 3. Depth of visor arc at center of back edge 2-7/8 (± 1/8) inches. 4. Distance from left tip to right tip of visor before assembly, when spread flat 8 (± 1/8) inches.					
4.	Inner band a. Overlap both ends of the high density polyethylene or the imitation (molded) polyethylene inner band whichever is applicable, 1 (± 1/8) inch and with the top and bottom edges even, stitch overlapped ends of inner band from top to bottom catching the inside end of inner band 1/8 (± 1/16) inch from the edge. OR As an alternate overlap both ends of the applicable inner band 1 (± 1/8) inch and secure with two staples. The first one approximately 3/8 inch down from top edge, the second one approximately 3/4 inch down from top edge, each catching both plies of the inner band.	101 or 301 or 401	LSa-1	7-8	A/3 A/3 A/3	 A/3 A/3

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Table I - Construction of cap frame (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
4.	Inner band (cont'd) The two staples shall be horizontally positioned and secured thru the solid (unperforated) area of the overlapped inner band. The flat head side of the staple shall be on the outside of inner band. b. The overlapped ends of the inner band shall be positioned 2 (± 1/4) inches to the left or right of center back. c. The imitation (molded) inner band, when applicable and patterned on one side only, shall finish with the pattern side on the outside.					
5.	Crown protector a. Fold one end of the crown protector on itself, back to back, and stitch 5/8 to 3/4 inch from the folded edge forming a loop. Each end of stitchline shall be backtacked. A double row of stitching may be used in place of backtacking.	101 or 301 or 401	EFa-1	8-10	A/3 A/3 A/3	 A/3 A/3
	b. Position and center the plastic card holder piece on the face side of the crown protector and stitch or heat seal around all four edges leaving a 1 1/2 to 1 5/8 inch opening on one side at the end furthest from the looped end of crown protector. Stitch a heat seal across the width of the card holder material 1 1/2 to to 1 5/8 inches down from the top edge forming an open end pocket on one side. The pocket opening may face towards either side of the cap frame.	304	LSbj-1 and SSV-1	6-8 (backtacking not required)	A/3	A/3
6.	Make welt strip a. Overedge stitch the entire top edge of the 2 inch wide wool welt strip.	500 series except 501	EFd-1	7-10	00/3	00/3

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Table I - Construction of cap frames (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
6.	Make welt strip (cont'd)					
	b. Join ends of welt strip 1/8 to 3/16 inch from raw edge, backtacking the ends of stitching. As an alternate to backtacking; TWO ROWS of superimposed stitching may be used.	101 or 301 or 401	SSa-1	12-14	A/3 A/3 A/3	 A/3 A/3
	c. Press seam open.					
	d. With the welt strip laid flat and seam face up, place a mark to indicate the center front.					
	e. Turn the welt strip so that the seam is on the inside.					
	f. Fold the welt strip lengthwise with the raw edge extending 1/2 (± 1/16) inch beyond the overedged edge. The width of the overedged portion shall be 3/4 (± 1/16) inch.					
	g. Form a permanent welt edge (sharp fold or tuck) in the lengthwise direction of the welt strip by stitching 1/16 inch from the foldline. The raised welt edge shall start on one side of the cap frame, approximately 5 inches back from front center of frame, continuing around the back of frame to the same position on the opposite side of frame.	101 or 301 or 401	OSe-1	12-14	A/3 A/3	 A/3
7.	Attach welt strip to inner band					
	a. Position the prepared welt strip over the entire bottom edge of inner band with the welt strip joining seam at center back of inner band, and the raised welt edge of the strip positioned 1/4 to 5/16 inch above the bottom edge of the band on the outside.					

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Table I - Construction of cap frames (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
7.	Attach welt strip to inner band (cont'd)					
	b. Stitch the welt strip to the entire circumference of the inner band adjacent to the stitchline used to form the raised welt edge. The height of this stitchline shall remain the same when stitching across the front of the frame, where a welt edge is not required.	101 or 301 or 401	BSa-1	8-10	A/3 24/4 24/4	A/3 or 30/3 A/3 or 30/3
	c. Stitch the top edge of the welt strip to the entire circumference of the inner band at the base of the overedge stitching.	101 or 301 or 401	LSa-1	8-10 (backtacking not required)	A/3 24/4 24/4	A/3 or 30/3 A/3 or 30/3
8.	Attach buttons (TYPE II FRAMES ONLY)					
	a. Punch a hole of proper diameter in each side of inner band and welt strip, with the center of hole 1/2 to 5/8 inch from bottom of inner band and 5 1/2 (± 1/8) inches front center of cap frame to center of each hole.					
	b. Insert securely a button with screw tube through each hole on each side of the inner band.					
9.	Prepare and attach front stand					
	a. Style A construction - Using the 2 inch wide by 6 1/2 inch long piece of white vinyl coated fabric cut in operation 1.g., bind the entire top edge of the molded front stand forming a 3/8 to 1/2 inch wide tunnel for inserting the grommet wire. The front free end of the crown protector piece shall be simultaneously folded under and caught in the stitching during this operation.	301 or 401	BSc-1	8-10	24/4 24/4	A/3 or 30/3 A/3 or 30/3

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Table I - Construction of cap frames (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
9.	Prepare and attach front stand (cont'd) b. Style B - construction - Center and drill two holes 3/16 (± 0) inches down from the top edge of the molded frontstand spaced 5/8 (± 0) inches apart. The diameter of each hole shall be 6/64 to 7/64 inches in diameter. c. Center the prepared molded front stand on the face side of the inner band for style A construction and on the back side of the inner band for style B construction, with the top center edge of the front stand positioned 2 1/4 to 2 1/2 inches above top edge of inner band. Secure bottom portion of the front stand to the inner band with six 1/2 inch wide staples properly positioned for maximum holding power. Staples of sufficient length shall be clinched horizontal to the inner band margins and shall finish smooth and flat on the backside of the inner band. Staple a support piece over the bottom portion of the frontstand on style A construction (see figure 3). The support piece may be made of any semi-flexible material considered suitable for the intended purpose. OR As an alternate, the lower portion of the front-stand may be stitched to the innerband with a minimum of two rows of 1 inch gage stitching. The top row of stitching shall be placed within the top margin of the innerband (see figure 3).	301	LSbj-2	5-7	24/4	24/4

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Table I - Construction of cap frames (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
9.	Prepare and attach front stand (cont'd) OR As an alternate, a combination of stapling and stitching may be used to attach lower portion of frontstand to inner band. c. The molded plastic front stand, before attachment to the frame, shall measure 6 (± 1/2) inches in width along top edge, 3 3/4 to 4 1/2 inches in width along bottom edge, and 3 3/8 to 4 inches in height at center front.					
10.	Attach visor Position the visor to the inside of inner band at center front and stitch 1/4 (± 1/16) inch from the skived edge of visor and 1/4 (± 1/16) inch from the bottom edge of the inner band.	301 or 401	LSa-1	5-6	24/4 24/4	A/3 or 30/3 A/3 or 30/3
11.	Make sweatband a. The flesh or backside of the sheepskin or artificial leather sweatband shall have a label affixed (see 3.3.17.1) approximately 1 inch to the right or left of the center back seam. The area to be labeled shall be unperforated. The tops of lettering shall face towards the top edge of sweatband.					
	b. Fold the 3/4 inch wide strip of plastic welting in half lengthwise. Position the bottom edge of sweatband 1/16 to 3/32 inch from the folded edge of welting and stitch thru all plies.	304	-	7-10	A/3 or 30/3	A/3 or 30/3

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Table I - Construction of cap frames (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
11.	Make sweatband (cont'd)					
	c. Abut the ends of sweatband assembly and join with the seam reinforced with a 5/8 inch wide strip of lightweight coated or uncoated fabric.	304	FSe-1	7-10	A/3 or 30/3	A/3 or 30/3
12.	Attach Sweatband					
	a. The back edge of visor shall be flanged upwards and set so as not to rub against the forehead when cap is worn.					
	b. Place the sweatband assembly on the inside of hat frame with folded edge of plastic welting positioned 1/8 to 3/16 inch above the bottom bound edge of inner band. The joining seam of sweatband shall be positioned at center back of frame.					
	c. Stitch sweatband assembly to the inner band with stitching going thru the plastic welting adjacent to the bottom edge of sweatband.	301	-	7-10	A/3 or 30/3	A/3 or 30/3
	d. Secure a size label (see 3.3.17.2) on the grain side of the sweatband positioned approximately 1 inch to the left or right of center back seam.					
13.	Attach grommet wire and crown protector					
	a. Clinch the adjustable steel clamp to one end of the properly cut size (length) of grommet wire.					
	b. Center and attach the properly sized length of grommet wire to top center of molded plastic front stand as follows:					

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Table I - Construction of cap frames (cont'd)

No.	Description of Operation	Stitch Type	Seam and Stitching Type	Stitches Per Inch	Thread	
					Needle	Bobbin or Looper
13.	Attach grommet wire and crown protector (cont'd) Style A - Thread free end of wire thru binding, crown protector loop and adjustment clamp at other end of wire. Overlap the wire ends approximately 1 1/2 inches and clinch the clamp. Style B - At a point one-half the distance of the wire's length, center and drill two holes 6/64 to 7/74 inches in diameter and spaced 5/8 (± 0) inches apart. Position wire along inside top edge of front stand and align the two sets of holes. Attach wire to front stand with two properly sized non-corrosive rivets, simultaneously catching the front end of the crown protector piece and rigid vinyl or non-corrosive metallic reinforcement piece on outside of frontstand. Thread wire thru crown protector loop and adjustment clamp. Overlap wires approximately 1 1/2 inches and clinch clamp.					
14.	Cleaning a. Trim ends of stitching and remove all loose thread. b. Remove all spots and stains.					

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3.9 Sizes and measurements - The sizes and measurements of the finished cap frames shall be in strict accordance with Table II. All measurements and tolerances are expressed in inches.

Table II - Sizes and finished measurements

Cap Frame Size	Head Circumference 1/	Length of Grommet Wire
6-1/2	20-1/2	34
6-5/8	20-7/8	34-3/8
6-3/4	21-1/4	34-3/4
6-7/8	21-5/8	35-1/8
7	22	35-1/2
7-1/8	22-3/8	35-7/8
7-1/4	22-3/4	36-1/4
7-3/8	23-1/8	36-5/8
7-1/2	23-1/2	37
7-5/8	23-7/8	37-3/8
7-3/4	24-1/4	37-5/8
7-7/8	24-5/8	38
8	25	38-3/8
Tolerance	+ 1/8, - 1/4	± 1/8

1/ The measurement shall be taken around the inside bottom edge of sweatband with an expandable type hat gauge.

3.10 Workmanship - The finished cap frames shall conform to the quality established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.1.1 Certificate of compliance - Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

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4.2 First article inspection - When required, the first article submitted in accordance with 3.2, shall be inspected as specified in 4.3.2.1 and 4.3.2.2 for compliance with design, construction, workmanship and dimensional requirements.

4.3 Quality conformance inspection - Inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated herein.

4.3.1 Component and material inspection - In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, figures, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document. In addition, testing shall be performed on the components specified in Table IV in accordance with FED-STD-191 wherever applicable, and as specified herein. The lot and sample size shall be in accordance with Table III and the lot shall be unacceptable if one or more sample units fail to meet any requirement specified. The units for expressing the lot sizes shall be as indicated in Table IV.

Table III - Sample size

Lot Size	Sample size
800 or less	2
801 thru 22,000	3
22,001 and over	5

Table IV - Component tests

Component and lot size expressed in terms of	Characteristic	Requirement paragraph	Test method
Front stand	Thickness	3.3.2	1/
	Stiffness	3.3.2	4.4.4
	Bending	3.3.2	4.4.5
Cloth, Cotton twill, visor (yds)	Material ident	3.3.3	1/
	Yarns per inch	3.3.3	1/
	Weight	3.3.3	1/
	Nonfibrous material	3.3.3	1/
Plumper paper, visor (yds)	Material ident.	3.3.4	1/
	Gauge	3.3.4	1/
	Immersion test	3.3.4	4.4.1

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Table IV - Component tests (cont'd)

Component and lot size expressed in terms of	Characteristic	Requirement paragraph	Test method
Plastic sheeting, visor (sheets or rolls)	Material ident.	3.3.5	1/
	Color	3.3.5	1/
	Stiffness and tackiness resistance	3.3.5	4.4.2
	Cracking and flaking resistance	3.3.5	4.4.3
Plastic visor, top (each)	Finish	3.3.5.1	1/
	Thickness	3.3.5.1	1/
Plastic visor, binding (feet)	Thickness	3.3.5.2	1/
	Width	3.3.5.2	1/
Leather, visor (pieces)	Material.	3.3.6	1/
	Finish	3.3.6	1/
	Tannage	3.3.6	1/
	Thickness	3.3.6	1/
Adhesive	Type	3.3.7	1/
	Water resistant	3.3.7	1/
Inner bands, standard and alternate (each)	Width	3.3.9	1/
	Thickness	3.3.9	1/
	Stiffness	3.3.9	4.4.4
	Bending	3.3.9	4.4.5
	Design	3.3.9	1/
Sweatband welting (yards)	Material	3.3.11	1/
	Thickness	3.3.11	1/
	Crack resistance	3.3.11	1/
Crown protector (sheets or rolls)	Material ident.	3.3.12	1/
	Weight	3.3.12	1/
	Crack resistance	3.3.12	1/
Card holder (sheets)	Material	3.3.13	1/
	Thickness	3.3.13	1/

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Table IV - Component tests (cont'd)

Component and lot size expressed in terms of	Characteristic	Requirement paragraph	Test method
Grommet wire (each)	Material ident.	3.3.14.1	1/
	Coating ident.	3.3.14.1	1/
	Width	3.3.14.1	1/
	Thickness	3.3.14.1	1/
Clamp (each)	Material ident.	3.3.14.2	1/
	Length	3.3.14.2	1/
Staples (Gross)	Material ident.	3.3.14.3	1/
	Thickness	3.3.14.3	1/
	Corrosion resistance	3.3.14.3	1/

1/ A certificate of compliance shall be submitted for this requriement.

4.3.2 Examination of the end item - The defects found during the examination of the end item shall be classified in accordance with 4.3.2.1 and 4.3.2.2. The applicable inspection levels and acceptable quality levels (AQL's) shall be as indicated in 4.3.2.3. The lot size shall be expressed in units of cap frames. The sample unit shall be one finished cap frame.

4.3.2.1 Visual examination

Defects	Classification		
	Major	Minor	
		A	B
I. Material Defects and Damages (General)			
a. Fabrics:			
- Hole, cut, tear, burn, or visible mend	X		
b. Leather:			
1. Hole, cut, tear, or crack	X		
2. Any evidence of the leather separating on the grain side		X	
3. Soft, spongy, or flanky		X	
4. Any scar, wrinkle or scratch		X	
5. Three or more exposed needle holes		X	
6. Three or more exposed vein marks		X	
7. Color not as specified		X	

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Defects	Classification		
	Major	Minor A	Minor B
c. Plastic materials:			
1. Hole, cut, or tear	X		
2. Color not as specified		X	
3. Evidence of separation, blistering, cracking or flaking		X	
II. Cleanness			
a. Any spot or stain		X	
b. Thread ends not trimmed or loose thread not removed throughout major portion of frame			X
III. Cutting			
-Any component part not cut in accordance with specification requirements	X		
IV. Component and Assembly			
a. Any component part or operation omitted	X		
b. Any component not as specified	X		
c. Any component part twisted or distorted		X	
d. Any operation not performed as specified		X	
V. Seams and Stitching			
a. Accuracy of seaming:			
1. End of stitching when not caught in another seam or stitching, backtacked less than 1/4 inch			X
2. End of a continuous line of stitching overlapped less than 1/2 inch			X
3. Thread breaks not secured or repair stitch starts less than 1/2 inch in back of the break			X
4. Stitching sewn too close to the edge resulting in damage to material	X		
5. Stitching irregular, not within range specified, or varies more than 1/16 inch when no range is specified		X	

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Defects	Classification		
	Major	Minor	
		A	B
V. Seams and stitching (cont'd)			
b. Open seams:			
1. Up to 1/4 inch, inclusive		X	
2. More than 1/4 inch	X		
NOTE: The seam shall be classified open when one or more stitches joining a seam are broken or when two or more continuous skipped or runoff stitches occur.			
c. Seam and stitch type:			
1. Not specified seam or stitch type		X	
2. Any line of stitching omitted		X	
3. Any line of stitching not beginning or ending where specified (unless otherwise classified herein)			X
4. Looper thread on outside (when 401 stitch type is used)		X	
d. Stitch tension:			
1. Loose tension resulting in a loose seam	X		
2. Tight tension (stitches break when normal strain is applied in the direction of the seam or stitching)	X		
e. Stitches per inch (score only when condition exists on more than 1/2 the length of seam):			
1. Less than the minimum specified		X	
2. More than the maximum specified			X
f. Overedge stitching:			
- Omitted for 1 or more inches on one or more seams		X	
VI. Molded front stand			
a. Offset from front center of cap frame			
1. Up thru 1/4 inch		X	
2. More than 1/4 inch	X		

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Defects	Classification		
	Major	Minor	
		A	B
VI. Molded front stand (cont'd)			
b. Top center edge less than 2 1/4 or more than 2 1/2 inches from top edge of inner band	X		
c. Less than 6 staples attaching stand to top margin of inner band		X	
d. Stand set on an angle, affecting appearance	X		
e. Staples not positioned for maximum holding power		X	
f. Staples not clinched smooth and flat		X	
g. Width across top edge of stand less than 5 1/2 or more than 6 1/2 inches		X	
h. Width of frontstand at top edge of inner band less than 4 3/4 inches		X	
i. Height of full stand at center front less than 3 3/8 or more than 4 inches		X	
VII. Grommet wire			
a. Clamp not securely clinched to wire		X	
b. Wire not cut to specified length	X		
c. Wire overlapped less than 1-3/8 inches		X	
d. Clamp does not function properly	X		
e. Rough or sharp edges on wire or clamp			X
f. Drill holes not properly positioned or sized, affecting serviceability (Style B)		X	
g. Wire cannot be threaded thru binding (Style A)	X		
VIII. Crown protector			
a. Twisted or poorly shaped		X	
b. Loop end loose or insecurely sewn			X
c. Loop end finished less than 5/8 or more than 3/4 inch		X	

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Defects	Classification		
	Major	Minor	
		A	B
VIII. Crown protector (cont'd)			
d. Not completely or not securely caught in steel wire or front stand binding affecting serviceability	X		
e. Too short, distorting shape of steel wire	X		
f. Too long, causing protector to sag excessively	X		
g. Face side of protector facing upwards		X	
IX. Card holder or crown protector			
a. Not stitched or heat sealed to face side of crown protector		X	
b. Not stitched or heat sealed on four sides, excluding open end of pocket		X	
c. Open end pocket:			
1. Less than 3-1/8 inches but not more than 3-3/8 inches in depth		X	
2. Less than 1-1/2 inches but not more than 1-5/8 inches in width		X	
X. Inner band			
a. Crushed, broken, warped or mishapen, affecting serviceability or appearance	X		
b. Overlapped ends insecurely fastened		X	
c. Molded design not facing outward	X		
d. Ends overlapped less than 7/8 inch or more than 1-1/8 inches		X	
e. One or both staples, when applicable, not horizontally positioned on the solid or unperforated portion of the inner band		X	
f. Holes for side buttons positioned less than 5 3/8 or more than 5 5/8 inches from the center of hole to center front of cap frame		X	

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Defects	Classification		
	Major	Minor	
		A	B
X. Inner band (cont'd)			
g. Holes for side buttons positioned less than 1/2 or more than 5/8 inch from bottom edge of inner band		X	
h. Holes for side buttons not proper diameter		X	
XI. Visor			
a. Not securely stitched to inner band	X		
b. Underside (leather) portion of visor not finished grain side up		X	
c. Distorted, twisted or puckered, affecting appearance	X		
d. Binding twisted, puckered, or pleated affecting appearance		X	
e. Binding not securely joined to edge of visor, affecting appearance		X	
f. Front edge poorly shaped, affecting appearance		X	
g. Placed off center with center of cap frame:			
1. More than 1/8 inch up to and including 1/4 inch		X	
2. More than 1/4 inch	X		
h. Placed off center with center of each button by 1/4 inch or more		X	
i. Width of visor at center less than 2 1/4 inches or more than 2 1/2 inches		X	
XII. Sweatband			
a. Exposed below bottom edge of cap frame or positioned more than 3/16 inch above bottom edge of cap frame		X	
b. Finished width less than 1 1/2 inches or more than 1 3/4 inches		X	

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Defects	Classification		
	Major	Minor A	Minor B
XII. Sweatband (cont'd)			
c. Twisted or puckered		X	
d. Color not as specified		X	
e. Back seam offset from center back of cap frame by more than 1/4 inch		X	
f. Ends not abutted		X	
g. Ends not securely stitched			X
XIII. Insignia (Type I, classes 1 and 2 visor top)			
a. Omitted	X		
b. Incorrect design	X		
c. Any defective area, affecting appearance		X	
XIV. Welt strip, wool			
a. Twisted, puckered, or pleated		X	
b. Irregular in width		X	
c. Too tight, distorting shape of innerband		X	
d. Raised welt edge omitted	X		
e. Raised welt edge less than 1/4 inch or more than 5/16 inch from bottom edge of inner band		X	
f. Raised welt edge irregular, affecting appearance		X	
g. Welt stitching not placed on upper side of raised edge		X	
h. Top edge not overedged stitched	X		
i. Row of attachment stitching at base of overedge stitching omitted	X		
j. Back seam of welt strip offset from center back of cap frame by more than 1/4 inch			X

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Defects	Classification		
	Major	Minor	
		A	B
XV. Labels			
a. Combination identification and size label:			
1. Omitted, incorrect, or illegible	X		
2. Misplaced		X	
b. Size label:			
1. Omitted, incorrect, or illegible	X		
2. Misplaced		X	
3. Insecurely attached		X	
c. Special instruction label:			
1. Omitted, incorrect, or illegible	X		
2. Size of label not as specified			X
3. Size of letters less than 3/32 inch in height			X
XVI. Buttons and screw tubes (when applicable)			
a. Length of screw tube not as specified		X	
b. Any defect on face of button, affecting appearance		X	
c. Damaged, does not function properly	X		

4.3.2.2 Size measurement - Any measurement deviating from dimensions and tolerances specified in Table II shall be scored as a size measurement defect.

4.3.2.3 Inspection levels and acceptable quality levels - The inspection levels and acceptable quality levels (AQL's), expressed in defects per 100 units (DHU), shall be as follows:

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	Inspection level	AQL
For visual examination in 4.3.2.1	III	
Major		1.5
Major and Minor A		6.5
Totals (Major, Minor A, and Minor B combined)		15.0
For measurement examination in 4.3.2.2	S-3	4.0

4.3.3 Examination of packaging requirements - An examination shall be made to determine that packaging, packing, and marking complies with the Section 5 requirements. Defects shall be scored in accordance with list below. The sample unit shall be one shipping container fully packaged for delivery with the exception that it need not be closed. Defects of closure listed below shall be examined on shipping containers fully packaged for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.
Material	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: incomplete closure of container flaps, loose strapping, inadequate stapling, or improper taping. Bulged or distorted container. Heat sealed seams open on polyethylene bags.
Content	Number of cap frames per shipping container is more or less than required. 1/ Size shown on one or more cap frames not as specified on shipping container. 1/

1/ For this defect, one shipping container in the sample shall be examined.

*4. Tests:

4.4.1 Immersion test - Two specimens of the plumper paper shall be immersed in tap water at 80°F for 15 minutes. The original shape shall be retained when dried.

4.4.2 Stiffness and tackiness test - A 2 by 4 inch test specimen of the unsupported plastic sheeting shall be doubled, face to face, placed between two glass plates under a pressure of one pound and heated for 30 minutes at 180°F in a thermostatically controlled oven. After allowing 5 minutes for cooling, the surfaces shall be unfolded and examined for any evidence of stiffening or tackiness. Two determinations per sample unit.

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4.4.3 Cracking and flaking test - A 2 by 4 inch specimen of the unsupported plastic sheeting shall be placed in a cold box at minus 20°F for 2 hours, and then tested at that temperature by folding face outward and creasing sharply. The plastic sheeting shall show no evidence of cracking or flaking. Two determinations per sample unit.

4.4.4 Stiffness test - The inner band material shall be tested for stiffness at room temperature according to FED-STD-311, method 4211, using a span length of 1.0 inch, a moment weight of 1.0 inch-pounds, and measuring the load at 20° on the angular deflection scale. The specimen shall be one inch wide by 4 inches long and it shall be cut from the center part of the band. Three determinations per sample unit.

4.4.5 Bending test - The test is run on a four inch long piece of inner band material. Each piece shall be bent across its width at least 180° around a metal test mandrel, a round metal bar measuring 1/8 inch in diameter. Each piece shall be examined for cracks, both while bent and after removing from the test. The test shall be run at 0°F. The specimen shall be straightened to its original position after removal from the mandrel. All samples and the test mandrel shall be kept in a cold box at a temperature of 0 ± 3°F for not less than four hours before the test. The test shall be run in the cold box and the samples and mandrel handled only with heavy gloves so as to prevent warming of the samples during the test. Two determinations per sample unit.

5. PACKAGING

5.1 Preservation-packaging - Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A - Each cap frame, with a special instruction label (see 3.3.17.3), shall be inserted into a clear polyethylene film bag having a thickness of 0.002 inch (± 20 percent tolerance). The bag shall be formed with heat sealed seams. As the alternate, the bag may be fabricated from polyethylene film tubing or sheeting. The cap visor shall face the bag opening and the opening of the bag shall extend a minimum of 6 inches beyond the cap visor. Closure of the polyethylene bag shall be accomplished by means of a mechanical tie (paper or plastic covered soft wire, aluminum band, etc.).

5.1.2 Level C (Commercial packaging) - Cap frames shall be packaged to afford adequate protection against physical damage during shipment from the contractor to the first receiving activity. The package and the quantity per package shall be the same as that normally used by the contractor for retail distribution.

5.2 Packing - Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A - Twelve cap frames of one type, class, and size only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC-L, type SF, grade V2s, class weather resistant of PPP-B-636. The inside of each container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade

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V15c of PPP-B-636. Level A packages shall be placed one on top of each other with alternate visors reversed front to back, two in length, one in width, and six in depth within a shipping container. Inside dimensions of each shipping container shall approximate 23-1/2 inches in length, 15 inches in width, and 15 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636. Toward the end of the contract or when there are less than the required amount per container of the same size, mixed sizes of one type and class only may be packed within the same container.

5.2.2 Level B - Twelve cap frames of one type, class, and size only, packaged as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC-L, type CF, variety SW or DW, class domestic, grade 275 of PPP-B-636. The inside of each container shall be fitted with box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. Level A packages shall be placed one on top of each other with alternate visors reversed front to back, two in length, one in width, and six in depth within a shipping container. Inside dimensions of each shipping container shall approximate 23-1/2 inches in length, 15 inches in width, and 15 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636. Toward the end of the contract or when there are less than the required amount per container of the same size, mixed sizes of one type and class only may be packed within the same container.

5.2.2.1 When specified (see 6.2), the fiberfoard shipping container shall be a grade V3c, W5c, or W6c fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of the container specification.

5.2.3 Level C (Commercial packing - Cap frames, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. The quantity per shipping container shall be the same as that normally used by the contractor for retail distribution. Containers shall comply with U.S. Postal Service Manual, Uniform Freight Classification Rules, or National Motor Freight Classification Rules, as applicable.

5.3 Marking - In addition to any special marking required by the contract, shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Labels, mixed sizes - Each shipping container packed with mixed sizes of one type and class only, shall have securely attached to the end and side, directly under the printing or stenciling, a white paper label 5 by 4 inches with the words "MIXED SIZES" plainly printed or stamped thereon and under these words shall be legibly printed or stamped the correct quantity and sizes contained therein.

6. NOTES

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6.1 Intended use - The cap frames covered by this specification are intended for use by male commissioned and non-commissioned personnel, midshipmen, aviation cadets, reserve officer training corps of the Navy and by all male personnel of the Coast Guard.

6.2 Ordering data - Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type, class, style, and size required (see 1.2).
c. Whether first article sample is required (see 3.2).
d. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
e. When weather-resistant grade fiberboard shipping containers are required for Level B packaging (see 5.2.2.1).

6.3 Samples - For access to samples, address the procuring activity issuing the invitation for bids.

6.4 Recycled material - It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification.

6.5 Metal fittings - The grommet wire and clamp described in 3.3.14 may be procured from Excel Tool and Die Company, 69 Sumner Street, Quincy, MA 02169. Any other supplier with a similar item that is equal to or better than the approved source will also be acceptable.

6.6 Metric conversion - The following multiplying factors are supplied for converting inch measurements found within this specification to the metric equivalent:

Inches to millimeters - Multiply inches by 25.4
Inches to centimeters - Multiply inches by 2.54

6.7 Changes from previous issue - The margins of this specification are marked with an asterisk to indicate when change (additions, modifications, corrections, deletions) from the previous issue made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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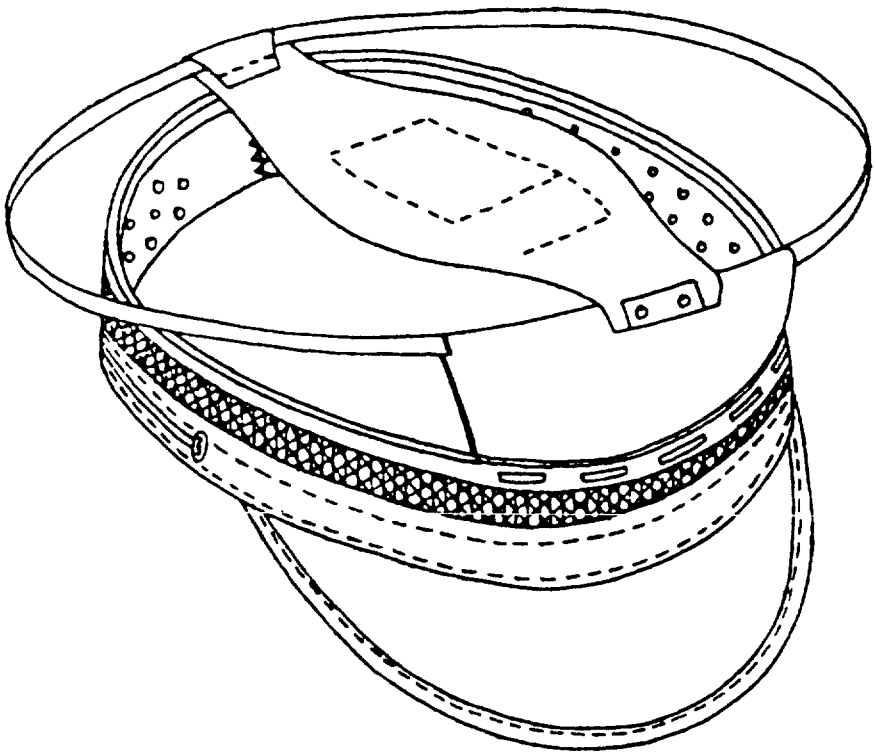
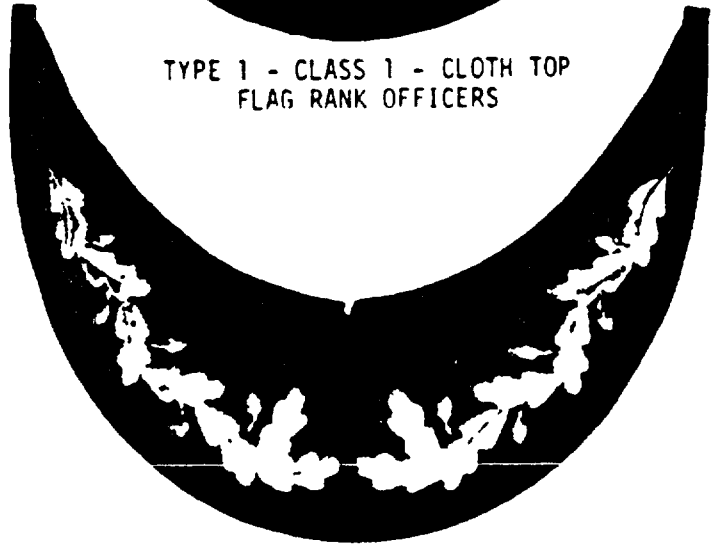


FIG 1 - FRAME, SERVICE CAP, MAN'S WITH STYLE B FRONT STAND

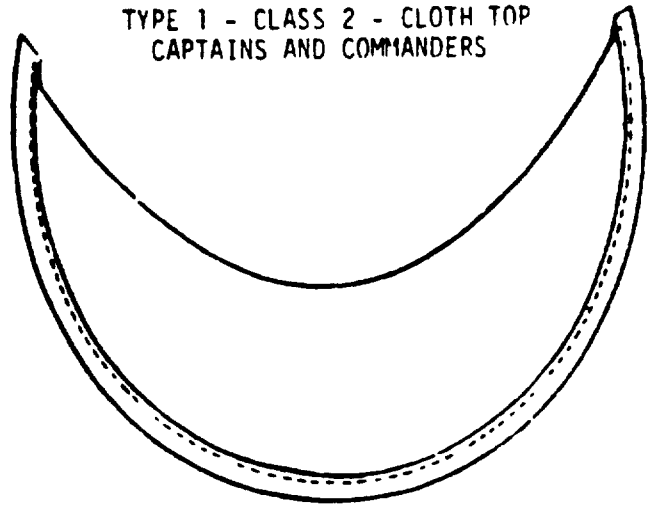
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TYPE 1 - CLASS 1 - CLOTH TOP
FLAG RANK OFFICERS



TYPE 1 - CLASS 2 - CLOTH TOP
CAPTAINS AND COMMANDERS



TYPE 2 - PLASTIC TOP

VISORS

FIGURE 2 - FRAME, SERVICE CAP, (MAN'S)

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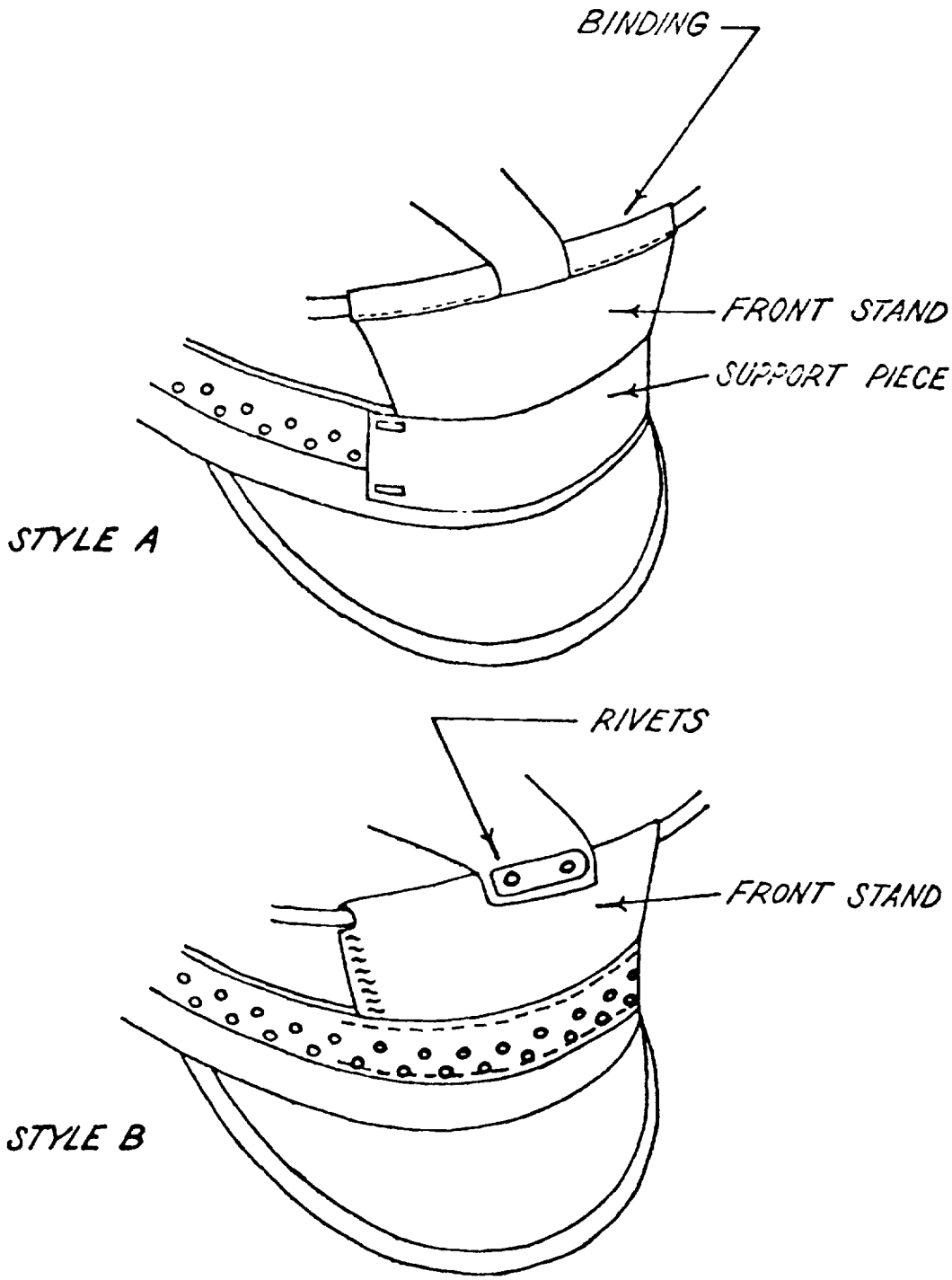
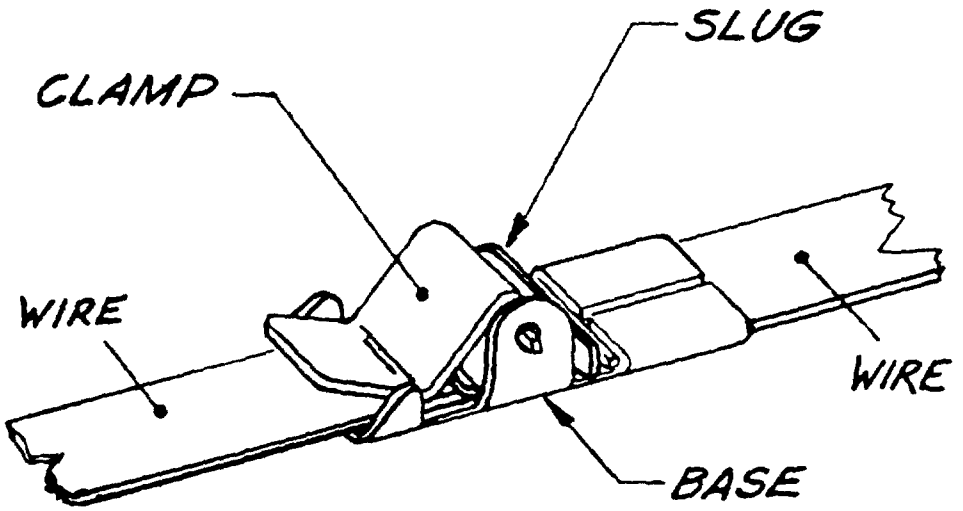
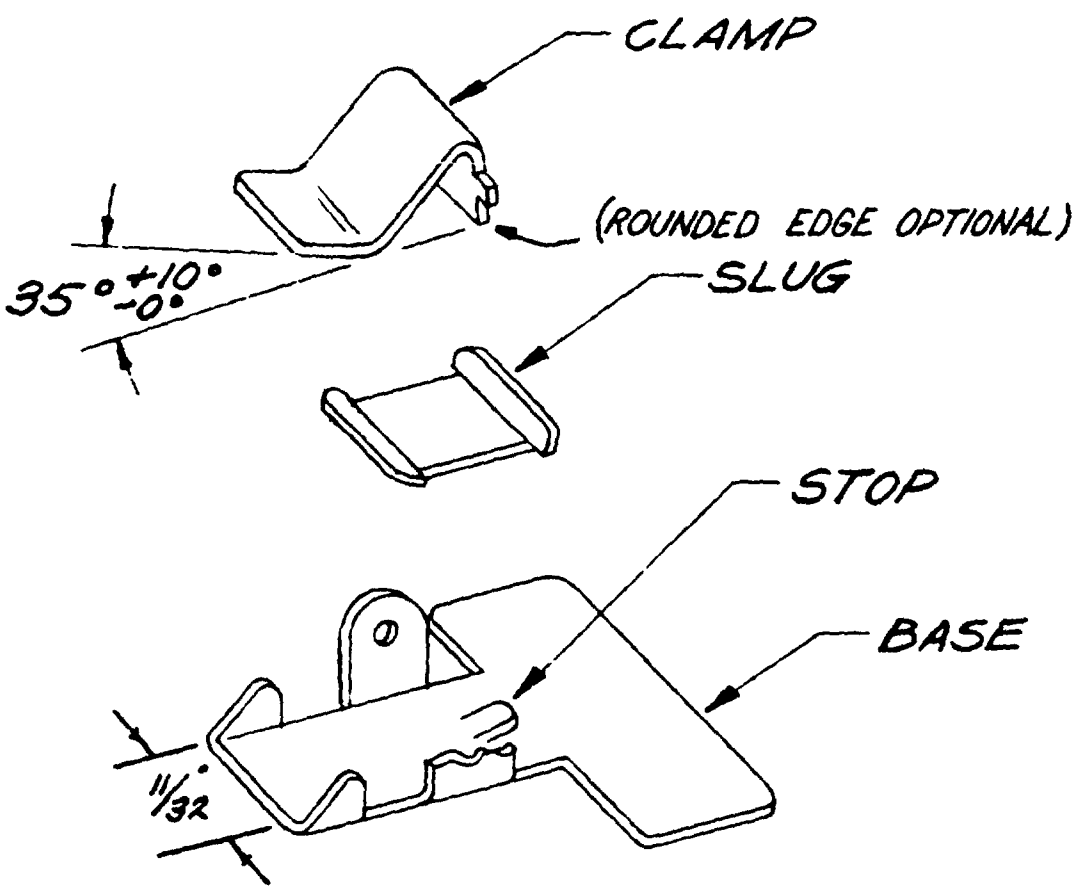


FIG 3- MOLDED PLASTIC FRONT STANDS

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ASSEMBLY



EXPLODED VIEW

FIG. 4 - CLAMP

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-F-17619E(NU) Frame, Service Cap (Man's)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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